CISC 473: Deep Learning — Midterm Report

P08: Abstractive Summarization via Parameter-Efficient Fine-Tuning

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Abstract

We investigate whether LoRA placement in BART-base's encoder self-attention, decoder self-attention, or encoder–decoder cross-attention affects summarization quality under matched trainable-parameter budgets. Each region adapts query and value projections at rank eight, yielding exactly 147,456 trainable parameters. We completed a single-seed pilot on a fixed 1,024-example CNN/DailyMail training subset for 32 optimizer updates and evaluated 64 held-out validation examples. Encoder, decoder, and cross-attention LoRA achieved ROUGE-1 scores of 41.02, 41.32, and 41.31, respectively, compared with 41.04 for unadapted BART-base. All three LoRA variants reduced validation loss, but generated-summary ROUGE changes were small and did not consistently improve across metrics. These preliminary measurements establish an operational, capacity-controlled comparison; they do not establish a reliable ranking of placements. Additional seeds, full fine-tuning and PEFT baselines, full test-set and cross-domain evaluation, and human factual-consistency assessment remain outstanding.

Introduction and Problem Statement

Abstractive summarization compresses lengthy source documents into fluent, salient, and factually accurate summaries. Pre-trained sequence-to-sequence Transformers such as BART [1] achieve high ROUGE scores by leveraging bidirectional encoder representations coupled with autoregressive decoder generation. However, conventional adaptation requires updating all 139.4 million parameters in BART-base, incurring steep computational overhead and high storage requirements when maintaining separate checkpoints for multiple domains or tasks. Parameter-efficient fine-tuning (PEFT) methods, particularly Low-Rank Adaptation (LoRA) [2], circumvent this cost by freezing the pre-trained weights $W_0 \in \mathbb{R}^{d\times k}$ and injecting trainable rank-decomposition matrices $\Delta W = BA$.

Midterm Scope and Rubric Alignment. This midterm presents zero-shot BART-base and the three isolated LoRA placements in a bounded pilot. The supplied P08 guide also requests full fine-tuning, Prefix Tuning, rank/projection ablations, cross-domain evaluation, and a human study. These requirements are not treated as waived. Their implementation or preparation is distinguished below from completed experiments; the pilot does not constitute published-baseline reproduction.

Research Question. Under strictly matched trainable parameter budgets and identical training conditions, how does the architectural placement of LoRA adapters (encoder self-attention vs. decoder self-attention vs. encoder-decoder cross-attention) govern summarization performance, factual consistency, and zero-shot cross-domain generalization?

Related Work (Updated)

Parameter-efficient fine-tuning for conditional text generation has evolved rapidly, transitioning from uniform parameter perturbation to structural and capacity-aware allocation.

Parameter-Efficient Sequence-to-Sequence Modeling. Large-scale pre-training of sequence-to-sequence architectures, exemplified by BART [1], established denoising objectives as the foundation for modern summarization. While full fine-tuning updates all parameters, Prefix-Tuning [3] introduced continuous virtual prefix vectors prepended to attention keys and values, preserving frozen model weights while achieving competitive generation on XSum with a small task-specific parameter budget. Concurrently, LoRA [2] demonstrated that task-specific parameter updates exhibit low intrinsic dimensionality, parametrizing weight updates via low-rank matrices $\Delta W = BA$. In their foundational work, Hu et al. observed that adapting attention query and value projections suffices for autoregressive language models. Its benchmarks include both language understanding and generation; our experiment specifically tests architectural regions within BART.

Non-Uniform Capacity Allocation and Layer Sensitivity. Recognizing that uniform low-rank parameterization across all layers may be sub-optimal, recent research has explored dynamic allocation. AdaLoRA [4] parameterizes weight matrices using singular value decomposition and iteratively prunes singular values based on importance scores, allocating higher rank to influential matrices. Similarly, WeightLoRA [7] evaluates adapter significance during fine-tuning, demonstrating that adapters are not uniformly useful throughout BART. In a broader theoretical analysis, He et al. [5] established a unified framework connecting adapters, prefix tuning, and LoRA, comparing the design choices shared by several PEFT methods. Furthermore, Valipour et al. [6] introduced DyLoRA to train models across a spectrum of ranks dynamically, supporting selection among multiple ranks without separately training every rank. These approaches motivate capacity-aware adaptation. Our study instead compares distinct, interpretable architectural regions under an identical, static adapter budget.

Factual Consistency in Abstractive Summarization. A known limitation of n-gram overlap metrics such as ROUGE-1, ROUGE-2, and ROUGE-L is their insensitivity to factual hallucinations. Kryscinski et al. [8] demonstrated that high-ROUGE summaries frequently introduce factual inconsistencies unsupported by the source text, proposing FactCC as a weakly supervised evaluation metric. Automatic factuality assessment also has domain limitations. We plan to complement ROUGE with a human study; neither a FactCC experiment nor human judgments are available in this repository.

Method	Year	Key Idea	Relation to Our Work
BART [1]	2020	Bidirectional encoder + autoregressive decoder seq2seq pre-training.	Primary backbone architecture for our summarization study.
LoRA [2]	2022	Low-rank weight update factorization ($\Delta W = BA$).	Core PEFT mechanism evaluated across architectural placements.
Prefix-Tuning [3]	2021	Virtual task-specific prefix vectors prepended to attention keys/values.	Baseline conditional conditioning approach for comparison.
AdaLoRA [4]	2023	SVD-based parameterization with dynamic rank allocation.	Demonstrates non-uniform capacity importance; motivates explicit placement study.
Unified PEFT [5]	2022	Unified functional framework linking adapters, prefixes, and LoRA.	Theoretical grounding for structural placement effects.
DyLoRA [6]	2023	Dynamic search of LoRA ranks during training.	Motivates explicit rank ablations as additional comparisons.
WeightLoRA [7]	2026	Selectable LoRA units pruned during training.	Confirms adapter utility varies across BART layers.
FactCC [8]	2020	Factual-consistency classifier for abstractive summarization.	Highlights ROUGE limitations and justifies our human evaluation protocol.

Table 1: Comparative taxonomy of related work and foundational methods.

Methodology

Our methodology enforces strict experimental consistency to reduce confounding when assessing whether performance differences originate from adapter placement rather than training variations or capacity disparities.

Model Architecture and Placement Isolation

We use pre-trained `facebook/bart-base` as our backbone, which comprises an encoder with 6 Transformer layers and a decoder with 6 Transformer layers ($d_{\text{model}} = 768$, 12 attention heads, feed-forward dimension $d_{\text{ff}} = 3072$, totaling 139,420,416 parameters). For any weight matrix $W_0 \in \mathbb{R}^{d \times k}$, LoRA parameterizes the forward pass as:

$$h = W_0 x + \Delta W x = W_0 x + \frac{\alpha}{r} BAx \tag{1}$$

where $A \in \mathbb{R}^{r \times k}$ uses PEFT's default Kaiming-uniform initialization, $B \in \mathbb{R}^{d \times r}$ is initialized to zero, r is the adaptation rank, and α is a scaling hyperparameter. We set $r = 8$ and $\alpha = 16$, targeting the query (W_q) and value (W_v) projections ($k = d = 768$). Each adapted projection introduces $2 \times 768 \times 8 = 12,288$ parameters.

We implement four LoRA configurations; the three isolated placements were executed in this pilot, while all-attention LoRA remains a comparison to run:

1. **Standard LoRA (All Attention):** Injects adapters into W_q and W_v across all 6 encoder self-attention, 6 decoder self-attention, and 6 decoder cross-attention blocks (18 attention layers

total $\times 24,576 = 442,368$ trainable parameters, 0.32% of the base model).

2. **Encoder Self-Attention LoRA:** Adapts only `model.encoder.layers.*.self_attn` ($6 \times 24,576 = 147,456$ trainable parameters, 0.11%).
3. **Decoder Self-Attention LoRA:** Adapts only `model.decoder.layers.*.self_attn` ($6 \times 24,576 = 147,456$ trainable parameters, 0.11%).
4. **Cross-Attention LoRA:** Adapts only `model.decoder.layers.*.encoder_attn` ($6 \times 24,576 = 147,456$ trainable parameters, 0.11%).

Crucially, configurations (2), (3), and (4) share an identical parameter budget of exactly 147,456 parameters, controlling adapter capacity in the architectural comparison.

Training Setup

The three adapted models use identical data membership, target projections, rank, seed, learning rate, and update budget. The completed pilot uses one Tesla T4 GPU with FP16, Python 3.12.12, PyTorch 2.5.1 (CUDA 12.4), Transformers 4.44.2, PEFT 0.13.2, Datasets 3.1.0, and Evaluate 0.4.3. The unadapted baseline freezes all model parameters and performs no optimization. Table 2 describes the *executed pilot*, rather than the longer default experiment schedule available in the code.

Hyperparameter	Executed pilot value
Backbone	`facebook/bart-base`, 139,420,416 base parameters
LoRA	Rank 8; $\alpha = 16$; dropout 0.05; query/value projections; no bias updates
Optimizer	AdamW, $\beta_1 = 0.9$, $\beta_2 = 0.999$, $\epsilon = 10^{-8}$
Learning rate	3×10^{-4} for each LoRA placement
Schedule	4 warmup updates, then linear decay; 32 optimizer updates total
Batch size	8 examples per device $\times$ 4 accumulation steps = 32 effective
Training exposure	One epoch over the same 1,024-example training subset
Regularization	Weight decay 0.01; gradient clipping at norm 1.0
Precision and hardware	FP16 on one Tesla T4
Generation	Beam width 4; length penalty 2; minimum length 56; maximum length 128
Repetition control	3-gram blocking; deterministic decoding without sampling
Randomness	Training seed 42 and data-selection seed 42; one run per configuration

Table 2: Matched settings for the completed single-seed placement pilot. Zero-shot BART-base is evaluated without training.

Dataset and Preprocessing

CNN/DailyMail version 3.0.0 has 287,113 training, 13,368 validation, and 11,490 test examples. We selected 1,024 training and 64 validation examples by shuffling each official split with data seed 42. The identical subsets are used across placement runs; zero-shot evaluation uses the same validation subset. The training and validation example-ID hashes in the exported manifests match across runs and were independently reproduced by reloading the pinned dataset. No test results are included in the evidence supplied for this report.

Sources truncate to 1,024 BPE tokens and training targets to 128 tokens. The tokenizer handles special tokens; dynamic label padding is masked with -100 in the loss. ROUGE uses the original,

untruncated reference summaries. We use the official Hugging Face Evaluate ROUGE implementation, with stemming, macro mean per-example F1 multiplied by 100, and ROUGE-L rather than ROUGE-Lsum. Validation loss uses truncated teacher-forced targets, so its improvement does not imply an improvement in generated-summary ROUGE or factual consistency.

The recorded model revision is `aadd2ab0ae0c8268c7c9693540e9904811f36177`; the CNN/DailyMail revision is `96df5e686bee6baa90b8bee7c28b81fa3fa6223d`. All runs record clean code revision `6c10d0a`, the same pinned metric revision, generation settings, and package environment. The final checkpoint is selected after the fixed update budget, without test-set selection. Full configurations and raw exported pilot records accompany the repository in `experiments/midterm_pilot_seed42`.

For future transfer evaluation, the code loads the same CNN/DailyMail-trained checkpoint on XSum without optimization. XSum has 204,045 training, 11,332 validation, and 11,334 test examples. No XSum metrics were supplied. Applying the current minimum generation length of 56 to single-sentence XSum summaries is a protocol choice to examine explicitly in subsequent evaluation.

Preliminary Results

Measured Budget-Matched Placement Comparison

The supplied Colab export contains four completed validation records: unadapted BART-base and the three isolated LoRA placements. Saved Trainer statistics are consistent with one epoch and 32 updates for each adapted model. Each placement targets 12 projections across six blocks and records exactly 147,456 trainable parameters. This is 0.1058% of the base-model parameter count, or 0.10565% of total parameters including adapters. Zero-shot inference has zero trainable parameters.

Configuration	Trainable	R-1	R-2	R-L	Val. loss	Loop s
Zero-shot BART-base	0	41.04	18.87	25.84	3.6372	0.0
Encoder self-attention	147,456	41.02	19.05	25.47	3.2659	135.4
Decoder self-attention	147,456	41.32	19.13	25.93	3.2647	105.5
Cross-attention	147,456	41.31	18.88	25.73	3.2401	103.7

Table 3: Single-seed pilot on 64 CNN/DailyMail validation examples (seed 42). LoRA runs use 1,024 training examples and 32 updates. Scores are ROUGE F1 $\times 100$; no standard deviations are available. Loop time includes epoch validation and checkpoint saving.

Observed Differences and Interpretation

Decoder self-attention has the highest observed ROUGE-1 (41.3206) and ROUGE-L (25.9262) in this sample; cross-attention has the lowest validation loss (3.2401). The decoder–cross ROUGE-1 difference is only 0.0094 points. The entire ROUGE-1 spread among the three placements is approximately 0.30 points. Relative to zero-shot ROUGE-1 (41.0442), decoder and cross-attention increase by approximately 0.28 and 0.27 points, while encoder-only adaptation changes by approximately -0.02 points.

The metric changes are mixed. Zero-shot ROUGE-L is 25.8363: decoder-only adaptation increases it by about 0.09 points, while cross-attention and encoder-only adaptation decrease it by about 0.11 and 0.36 points. All three adapted models reduce teacher-forced validation loss from the baseline's

3.6372 to approximately 3.24–3.27. The distinction between lower loss and mixed ROUGE changes reinforces the need to evaluate actual generated text rather than relying on loss alone.

With one seed, 64 validation examples, and short training, these observations do not identify a statistically reliable best placement or establish convergence. No standard deviations, confidence intervals, or significance claims are reported. The pilot does not support a claim that cross-attention consistently outperforms decoder or encoder adaptation, nor a claim about factuality or transfer.

Timing and Evidence Boundaries

Recorded training-loop wall times are 135.4 seconds for encoder self-attention, 105.5 seconds for decoder self-attention, and 103.7 seconds for cross-attention. These measurements include the Trainer's epoch validation and checkpoint saving; they exclude initial model/data preparation and the separately timed final validation prediction pass. The latter takes approximately 50.5–57.2 seconds for the LoRA runs. Zero-shot's zero training time means no optimization was performed, not that inference was free: its validation took 67.5 seconds. These single-run times do not establish a robust efficiency ranking or equal wall-clock/FLOP budgets.

The raw uploaded JSON and Markdown table agree at their reported precision. Configuration, target-module, parameter-count, environment, data-hash, and metric checks establish internal consistency of the supplied records. Predictions and checkpoints were not uploaded, so ROUGE values were not independently recomputed. The exported aggregate scores provide no per-example factuality assessment.

Remaining Baselines, Ablations, and Evaluation

Full fine-tuning, Prefix Tuning, and all-attention LoRA are implemented comparison options, but no completed research results for them were supplied. Rank $r \in \{4, 8, 16\}$ and Q/K/V/all-projection ablations remain to be run. Seeds 123 and 999, longer training, and full held-out CNN/DailyMail and XSum evaluations are outstanding. Even if optional subset test evaluations were run in Colab, their files were not included in the current evidence and are not claimed.

The guide's baseline-reproduction and multi-seed criteria are not yet satisfied. We will compare mean and sample standard deviation after collecting at least three independent runs. Factual consistency additionally requires the planned 10-participant, 20-summary human study; no human or FactCC results are available.

Challenges and Adaptations

The original training scaffold did not connect ROUGE to the Trainer, and its fallback evaluator substituted invalid approximate scores. We replaced this with pinned official Evaluate ROUGE, explicit failure propagation, and saved configuration/metric/prediction artifacts. Shared model construction now checks actual PEFT targets and trainable counts; one evaluation-only path supports both test domains. CPU training, generation, and checkpoint-reload smoke tests were executed; the documented suite passed 29 tests before the pilot.

Colab's newer notebook Python initially conflicted with the pinned experiment stack. The setup now creates a separate Python 3.12.12 environment. Under the midterm deadline, we used a bounded single-seed pilot with a 1,024-example training subset and 64 validation examples, preserving matched placement budgets and settings. This adaptation supplies genuine preliminary

measurements but limits what can be concluded and does not replace the remaining rubric requirements.

A result-validation script checks the uploaded records, reproduces the report table and single-seed parameter plot, and rejects incompatible or missing evidence. Remaining work prioritizes comparison baselines, additional seeds, held-out transfer, and factuality evaluation, rather than interpreting the current small ROUGE differences as definitive.

Updated Week 7–10 Plan

Table 4 outlines our roadmap for the remaining project period.

Week	Tasks	Responsible
7	Complete full fine-tuning, Prefix Tuning, and all-attention baselines; repeat placements with seeds 123 and 999; extend training and validate stability.	Wen Bin / Junn Rong
8	Evaluate full CNN/DailyMail and XSum test splits without transfer fine-tuning; run required rank/projection ablations; conduct the 10-participant, 20-summary human study.	Junn Rong / Wen Bin
9	Synthesize ROUGE, factuality, and efficiency results; generate parameter-performance Pareto trade-off curves; conduct qualitative failure case analysis.	Both
10	Final report drafting, comprehensive code clean-up, reproduction verification from a clean environment, and presentation deck preparation. **(Final due)**	Both

Table 4: Updated project schedule and task allocation for Weeks 7–10.

Preliminary Novel Contribution (Optional Bonus)

Our extension is a controlled comparison of three interpretable attention regions under an exact adapter budget. The completed pilot demonstrates that this comparison is operational and reproducible from saved configurations, and it exposes differences between loss and generation metrics. It does not establish superior quality, efficiency, transfer, or factuality for any placement. Stronger empirical or novelty claims require the outstanding baselines, multiple seeds, broader evaluation, and continued comparison with prior work.

Contribution Declaration

The team declares equal contributions: Choo Wen Bin, 50%, and Nai Junn Rong, 50%. These author-provided percentages are retained; the supplied experiment records do not establish individual task ownership. The team should confirm actual responsibilities before submission.

AI Usage Disclosure

An OpenAI Codex assistant was used for this code audit, implementation corrections, test and documentation preparation, Colab setup troubleshooting, result-consistency checks, and drafting/revising the report. The project team executed the Colab pilot and supplied its saved outputs.

Reported pilot measurements are transcribed from those files; AI-generated code and prose require author review. No missing measurements or human judgments were fabricated. This disclosure covers the current revision; the team must disclose any earlier AI use separately and confirm the final wording before submission.

References

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